Multi-Graph Contextual Learning for Multimodal Meme Classification

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ABSTRACT

This work proposes a **multi-graph contextual classification framework for multimodal meme analysis** that leverages relationships between samples rather than treating them independently. Experiments are conducted on the **What's Beneath Misogynous Stereotyping (WBMS) Dataset**, which contains meme images and captions annotated for misogynistic content. Text and image features are extracted using **BERT** and **CLIP**, respectively, and reduced using Principal Component Analysis (**PCA**). Three similarity graphs are then constructed using cosine similarity–based k-nearest neighbors (**kNN**): a text semantic graph, a visual semantic graph, and a class-discriminative graph obtained using Linear Discriminant Analysis (**LDA**). These graphs are processed using Graph Attention Networks (**GAT**) to learn contextual node representations through attention-based neighbor aggregation. The contextual features from the graphs are fused to perform final classification. This approach aims to incorporate dataset-level contextual information by modeling statistical relationships between samples in the embedding space.

Keywords: Multimodal Meme Classification, Graph Neural Networks, Graph Attention Networks

Literature Review

Paper	Key Idea	Method	Contribution
KNN-GNN: Enhancing GNNs via Common Subspace	Improve GNN performance when graph connections do not reflect feature similarity	Learn a shared latent subspace and retrieve feature-similar neighbors using kNN	Enables aggregation from non-local but similar nodes; improves representation learning
kNNGNN: Learning kNN Hyperparameters via GNNs	Reformulate kNN as a learnable graph-based model	Construct micro-graphs with query node and neighbors and apply Message Passing Neural Network (MPNN)	Learns similarity metrics and neighbor weights automatically instead of manual tuning
kNN-KGE: Retrieval-Augmented Knowledge Graph Embedding	Use nearest neighbor retrieval to improve knowledge graph reasoning	Retrieve k nearest entities from a knowledge store using embedding similarity	Improves prediction for rare entities using external contextual memory

Proposed Solution

Multi-Graph Contextual Learning for Multimodal Meme Classification

Step 1 – Feature Extraction

- Text embeddings: BERT
- Image embeddings: CLIP
- Dimensionality reduction using PCA

Step 2 – Graph Construction

Three similarity graphs using cosine similarity and KNN:

- Text Semantic Graph
- Visual Semantic Graph
- Class-Discriminative Graph (LDA projection)

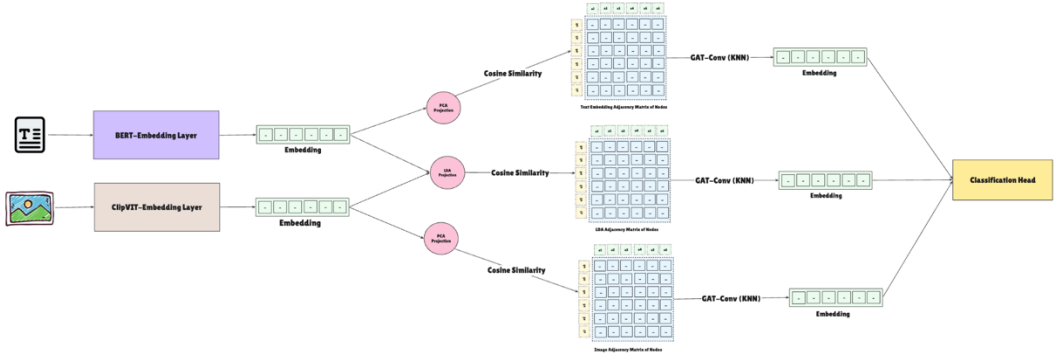
Step 3 – Contextual Learning

Graph Attention Networks aggregate neighbor information:

$h_i' = \sigma(\Sigma_j \alpha_{ij} W h_j)$

Step 4 – Feature Fusion

Contextual representations from all graphs are fused for final classification.



Preliminary Results

Preliminary experiments were conducted to evaluate the effect of incorporating class-semantic relationships before implementing the full multi-graph architecture. At this stage, the modality-specific graphs (text and visual graphs) were not included. Instead, the focus was on analyzing the impact of the class-semantic graph constructed in the LDA space.

Table 1: Performance Comparison of Models

Model	Split	Accuracy	Precision	Recall	F1 Macro	F1 Weighted
Graph	Train	0.9847	0.9847	0.9847	0.9709	0.9846
	Test	0.9554	0.9550	0.9554	0.9247	0.9543
No Graph	Train	0.9425	0.9410	0.9425	0.8869	0.9395
	Test	0.9061	0.9038	0.9061	0.8144	0.8977
PCA Only	Train	0.9982	0.9983	0.9982	0.9968	0.9982
	Test	0.9507	0.9490	0.9507	0.9099	0.9491

Future Scope

Completion of Multi-Graph Framework

- Integrate the **Text Semantic Graph** and **Visual Semantic Graph** with the current class-semantic graph.
- Perform joint contextual learning across all three graphs.

Improved Graph Fusion

- Explore advanced **feature fusion strategies** for combining contextual representations from multiple graphs.

Model Evaluation

- Conduct extensive experiments on the **WBMS dataset** using standard classification metrics.
- Perform ablation studies to evaluate the contribution of each graph component.

Generalization to Other Multimodal Tasks

- Apply the framework to other multimodal datasets such as meme sentiment analysis or multimodal misinformation detection.

Thank You